Network Threats and Mitigation & Physical Security and Risk

Q1. Create a table listing 5 common network threats (such as phishing, DDoS, man-in-the-middle, malware, and password attacks), and for each, write one real-world example from an app or service you use (like WhatsApp, Instagram, or Paytm).

Ans:

Network Threat	Description	Real-World Example
Phishing	Fraudulent emails, messages, or websites designed to steal user credentials or personal information.	A fake **Paytm** login page asking users to enter their username, password, or OTP.
Distributed Denial-of-Service (DDoS)	An attack that floods a server with excessive traffic, making the service unavailable to legitimate users.	**Instagram** becomes temporarily unavailable due to a large-scale DDoS attack.
Man-in-the-Middle (MITM)	An attacker secretly intercepts communication between two parties to steal or modify data.	A hacker intercepts **WhatsApp** traffic on an unsecured public Wi-Fi network.
Malware	Malicious software that damages systems, steals data, or disrupts normal operations.	A fake **WhatsApp APK** installs spyware that steals personal information.
Password Attack	Attackers attempt to gain unauthorized access by guessing or stealing user passwords.	A brute-force attack targeting an **Instagram** account with a weak password.

Conclusion

Network threats such as phishing, DDoS attacks, man-in-the-middle attacks, malware, and password attacks can compromise user data and disrupt online services. Understanding these threats helps organizations implement effective security measures to protect their networks and users.

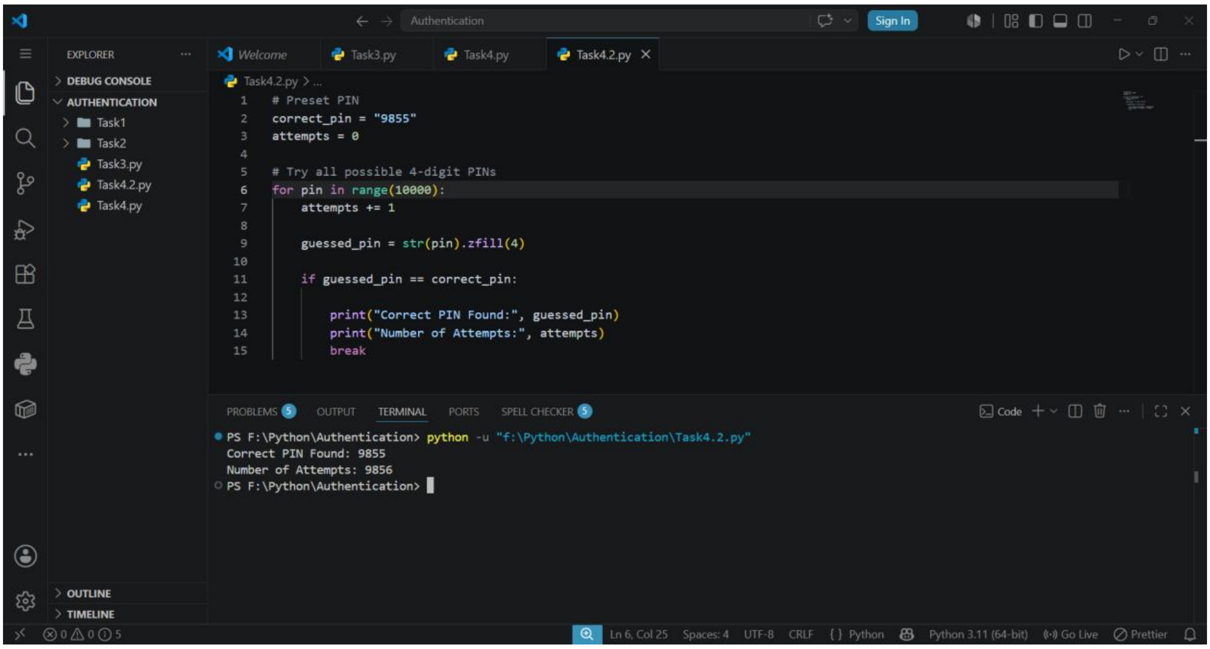
Q2. Simulate a brute-force attack scenario by writing a simple Python script that tries to guess a 4-digit PIN code by checking all possible combinations against a preset value. The script should print the number of attempts it took to guess the correct PIN.

Ans:

Algorithm:

1. Store a preset 4-digit PIN in a variable.
2. Initialize a counter to track the number of attempts.
3. Use a **for loop** to generate all possible 4-digit PIN combinations (0000–9999).
4. Convert each number into a 4-digit string using zfill(4).
5. Compare each generated PIN with the preset PIN.
6. If the PIN matches, display the PIN and the total number of attempts.
7. Stop the program.

Figure 1: Brute-Force Attack Simulation Program and its Output



Explanation

The program simulates a brute-force attack by systematically trying every possible 4-digit PIN combination from **0000** to **9999**. Each generated PIN is compared with the preset PIN until a match is found.

A counter records the number of attempts made. When the correct PIN is identified, the program displays the matching PIN along with the total number of attempts required.

Conclusion

The Python program successfully demonstrates how a brute-force attack works by checking all possible PIN combinations until the correct one is found. It also highlights why strong passwords, account lockout mechanisms, and multi-factor authentication are important for protecting user accounts.

Q3. List 4 physical security risks that could affect a server room hosting a popular app like Zomato or BookMyShow, and suggest one mitigation step for each risk.

Ans:

Physical Security Risk	**Real-World Example**	**Mitigation Step**
Unauthorized Physical Access	An unauthorized person enters the server room and tampers with network equipment.	Install biometric access control and allow entry only to authorized personnel.
Fire Hazard	An electrical short circuit causes a fire, damaging servers and network devices.	Install smoke detectors and automatic fire suppression systems.
Power Failure	A power outage shuts down the server room, making the Zomato or BookMyShow service unavailable.	Use UPS systems and backup generators to provide continuous power.
Equipment Theft or Tampering	A server or networking device is stolen or physically damaged by an intruder.	Install CCTV cameras, secure server racks with locks, and monitor the server room continuously.
Overheating	Cooling system failure causes servers to overheat, resulting in hardware failure and service interruption.	Install air conditioning, temperature sensors, and proper ventilation to maintain an optimal server room temperature.

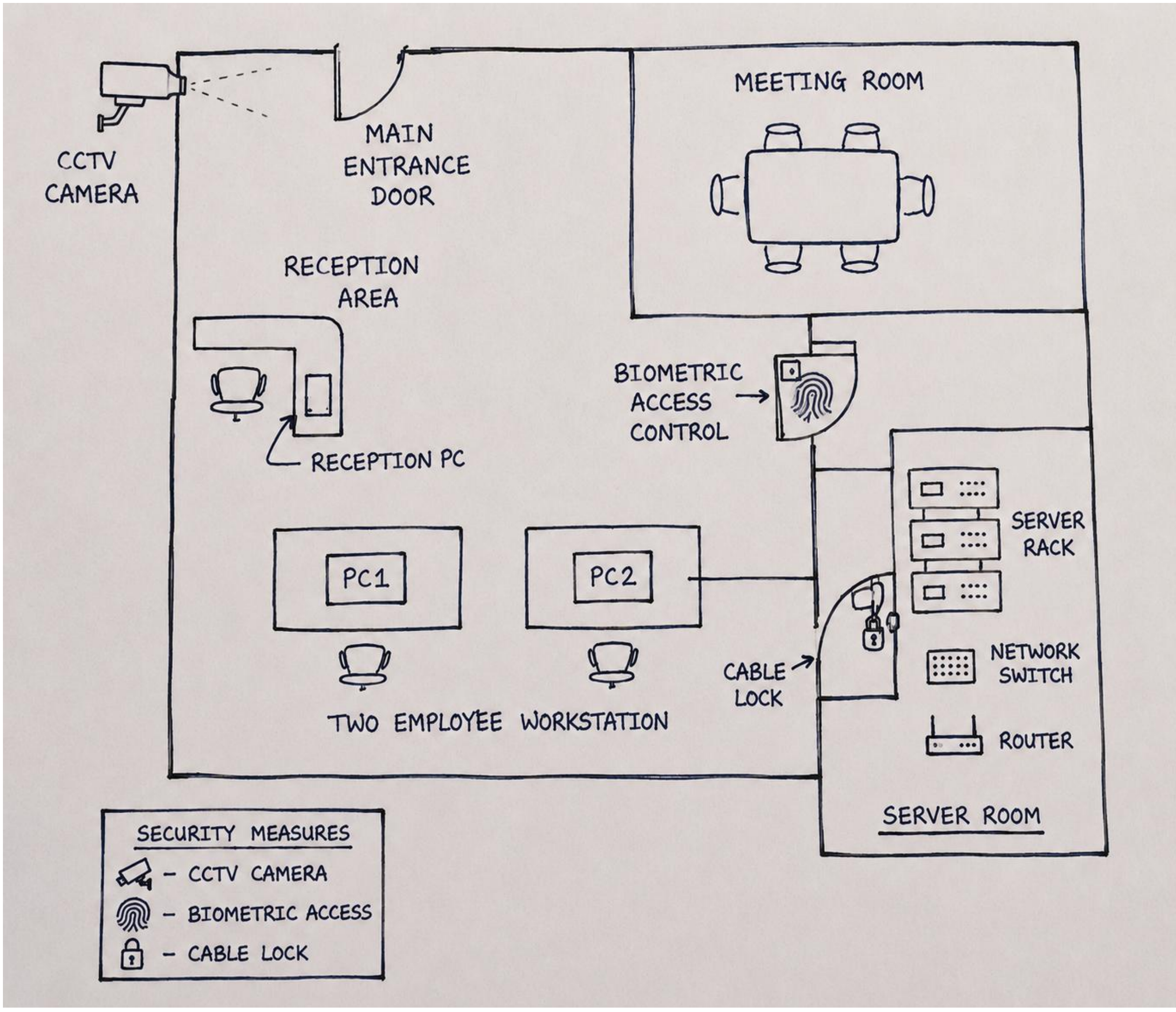
Conclusion

Physical security is essential for protecting servers and networking equipment from risks such as unauthorized access, fire, power failure, theft, and overheating. Implementing appropriate security measures helps ensure continuous, reliable, and secure operation of services such as Zomato and BookMyShow.

Q4. Draw a simple floor plan (hand-drawn or digital) of a small office and mark at least 3 physical security measures you would implement to protect network equipment (such as CCTV, biometric access, or cable locks).

Ans:

Figure 1: Floor Plan of a Small Office with Physical Security Measures



Physical Security Measures Used

Security Measure	Purpose
CCTV Camera	Monitors the office and server room to detect suspicious activities.
Biometric Access	Allows only authorized employees to enter the server room.
Cable Lock	Prevents theft or unauthorized removal of network equipment.

Q5. Use ChatGPT or a similar AI tool to generate a checklist of 7 actions a network admin should take to mitigate risks from both network threats and physical security breaches, then review and edit the checklist to make it specific for a college computer lab environment.

Ans:

Checklist Generated by ChatGPT

No.	Action	Why It Is Important
1	Use strong passwords and change them regularly.	Prevents unauthorized access.
2	Keep operating systems and antivirus updated.	Protects against malware and security vulnerabilities.
3	Enable firewalls and monitor network traffic.	Blocks suspicious network connections.
4	Take regular backups of important data.	Prevents data loss.
5	Restrict physical access to network devices.	Prevents theft and tampering.
6	Disable unused ports and services.	Reduces security risks.
7	Monitor security logs regularly.	Helps detect suspicious activities quickly.

Reviewed and Edited for a College Computer Lab

After reviewing the checklist, the following changes were made to make it suitable for a college computer lab:

- Students should log in only with their assigned accounts.

- The lab administrator should update all computers and antivirus software regularly.
- Routers, switches, and servers should be kept in a locked server room.
- CCTV cameras should monitor the computer lab.
- USB devices should be allowed only with administrator permission.
- Important lab files should be backed up regularly.
- The lab administrator should regularly check firewall and network logs.

Conclusion

The checklist was reviewed and modified to meet the security needs of a college computer lab. These measures help protect computers, network devices, and student data from both cyber threats and physical security risks.